

Aniket Arora

Senior Data Scientist | ML, Analytics & GenAI Enablement

Brussels, Belgium

arora.aniket@hotmail.com | aniketarora.be | linkedin.com/in/aniket-arora

github.com/AniketArora | x.com/AniketArora_1

Summary

Senior Data Scientist at IBM Consulting in Energy & Utilities, delivering predictive maintenance, KPI analytics, and agentic forecasting solutions. Built reusable Python analytics foundations that reduced extraction time 3–5x (from 60+ minutes to 10–15 minutes) and enabled faster intervention prioritization for maintenance teams.

Experience

IBM Consulting

Brussels, Belgium

Industry focus: Energy & Utilities

Senior Data Scientist

Jan 2025 – Present

- Architected and built a Python ETL foundation across Elasticsearch, Postgres, and TimescaleDB; standardized cleaning/type validation, duplicate and null handling, and DataFrame merges while introducing uv, pre-commit, and ruff for maintainable analytics code.
- Contributing to an agentic system for financial forecasting and simulations, supporting AI-driven scenario analysis and forecasting workflows.
- Supporting a financial forecasting model built with Facebook Prophet and working across Snowflake and Azure, including Azure Machine Learning (Azure ML) and Azure Data Factory (ADF).
- Reduced extraction time 3–5x (from 60+ minutes to 10–15 minutes) and saved ~2–3 hours per analysis cycle over 3+ months.
- Identified a vendor software issue affecting ~50% of devices; supported a patch that resolved ~99% of impacted units.
- Drove predictive maintenance analyses for a multi-million device portfolio using ensemble models (XGBoost), enabling faster intervention prioritization and supporting a substantial reduction in incident share.
- Led resolution of ~200 incidents per month with a small team; translated analytics into actionable remediation plans with clients and partners.
- Engineered high-performance KPI pipelines using SQL and TimescaleDB; optimized query performance via materialized views, reducing reporting latency from hours to ~3 minutes and cutting query runtimes from minutes to seconds.
- Established analytics standards and onboarded 6–7 maintenance team members on project setup, business context, investigation scripts, and ITIL processes.

Data Scientist

Apr 2022 – Dec 2024

- Built an end-to-end geospatial analytics workflow using cell tower data and Plotly to identify device clusters impacted by telecom network issues.
- Developed reusable investigation notebooks adopted by a 6–7 member maintenance team and still used for recurring triage and root-cause analysis.
- Partnered with stakeholders to define operational KPIs and translate analytics into business-facing reporting and decision support.

Junior DevOps Engineer

Sep 2021 – Apr 2022

- Improved a client ecosystem on GitLab; established Angular best-practice guidelines and built a PoC to raise development quality.

Automation Intern

Feb 2021 – Sep 2021

- Implemented internal automations and evaluated RPA vendors to streamline workflows; produced documentation and training materials to support enterprise-wide adoption.

Selected Projects (Personal)

Apollo Fitness (AI Agent Ecosystem) | Python, LangChain, PyTorch, OpenAI API

2024 – Present

- Developing a fitness ecosystem using AI agents to generate personalized workout and nutrition plans.
- Implemented Agentic workflows and RAG to process user preferences and provide real-time, context-aware suggestions.

- Demoed a private prototype to a small group of testers.

Government Client Hackathon (3rd place) | IBM watsonx

2025

- Built an AI assistant prototype for dispute-resolution advisory in a 3-day multidisciplinary hackathon.

Office of the Future (Client: ML6) | OpenCV, Google Cloud, Flask

2021

- Built a computer-vision prototype to analyze office occupancy and space utilization.

Talks & Teaching

- Guest Lecturer, Howest University of Applied Sciences (2025) – Delivered sessions on RAG, AI Agents, and Responsible AI for student and industry audiences.

Skills

AI & Machine Learning

Generative AI, LLMs, Agentic Workflows, RAG, PyTorch, TensorFlow, LangChain, Scikit-learn, XGBoost

Data & Cloud

Python (Pandas, NumPy), SQL, Elasticsearch, Postgres, TimescaleDB, Snowflake, Azure, Azure Machine Learning (Azure ML), Azure Data Factory (ADF), Databricks (upcoming), Grafana, Docker, Google Cloud Platform (GCP), IBM Cloud, IBM watsonx, Git

Languages

Dutch (Native), English (Fluent), Hindi (Conversational), French (Elementary)

Certifications

- **IBM Generative & Agentic AI Foundation** (2024)
- **IBM AI Associate Data Scientist** (2022)
- **Energy, Environment and Utilities Insights and Solutions (Bronze)** (2025)
- **Trustworthy AI and AI Ethics** (2022)
- **Enterprise Design Thinking Practitioner** (2024)

Education

Howest

2018 – 2021

Bachelor's degree, New Media & Communication Technology (NMCT)

Thesis: In what situation can I use Go in an MLOps pipeline compared to Python?